The Scarring Effects of Workplace Sexual Harassment

Nisha Chikhale, Natalie Duncombe, and Birthe Larsen*

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Abstract

We document new facts about the sources and consequences of sexual harassment in the workplace using administrative and survey data from Denmark. We estimate that following workplace sexual harassment, victims, both men and women, see earnings losses of around 6 percent. Losses are largely driven by workers who transition to new firms following harassment suggesting that the consequences of harassment persist even when workers leave the job where the harassment occurred. Victims who leave their employer after an incident of workplace sexual harassment incident move to firms with a larger share of colleagues who share their own gender.

*Chikhale: University of Delaware, Duncombe: U.S. Government Accountability Office, Larsen: Copenhagen Business School. We thank numerous seminar participants for helpful comments. The findings and conclusions of this paper are those of the authors and are not purported to represent the views of the U.S. Government Accountability Office.

1 Introduction

Workplace sexual harassment is a pervasive problem facing many workers. Often, the only way to escape workplace sexual harassment is to leave your job (Feldblum and Lipnic, 2016). Although workplace sexual harassment can affect all workers, the rate, and thus the burden, of sexual harassment is much higher for women than men.[1] Despite the prevalence of workplace sexual harassment, not enough is known about the earnings consequences for victims.

The aim of this paper is to address this gap. We begin our analysis using a data set that links administrative and survey data from Denmark. The survey provides us with self-reported experiences of workplace sexual harassment that we match with workers' labor market histories. The data allow us to estimate the earnings consequences of workplace sexual harassment. We find that the earnings scars of workplace sexual harassment can be significant and persist even when victims are removed from the site of harassment. Earnings losses are twice as high for victims who "escape" workplace sexual harassment and move to a new firm.

We start our analysis with a reduced-form evaluation of the earnings consequences of workplace sexual harassment. We use survey data provided by the Danish National Research Center for the Working Environment which asks workers about the frequency and sources of sexual harassment in their workplace. One contribution of our work, is to provide direct evidence on who perpetuates workplace sexual harassment. Consistent with predictions of prior work, coworkers are a significant source of workplace sexual harassment. However, we also document a large proportion of incidences stemming from clients, particularly as a source of harassment for women.

We employ an event study approach to estimate the consequences of workplace sexual harassment. We find that on average workplace sexual harassment is followed by earnings losses of 5-6 percent for both men and women. This number doubles for workers who leave

[1] Our data contain a binary measure of gender, preventing us from identifying if an affected worker is gender non-conforming/non-binary or differentiating between cisgender and transgender workers. Workers in gender minority groups are likely to face higher rates of harassment due to their marginalized gender. However, we are unable to speak to this with existing data.

their firm indicating that the earnings scars of workplace sexual harassment persist even when workers escape the reach of their harasser. Looking at workers who move firms following workplace sexual harassment, we see that men and women tend to move to firms with a larger share of workers who share their gender, and thus potentially face a lower risk of continued sexual harassment.

Work documenting the prevalence and consequences of workplace sexual harassment has a longer history in the fields of sociology, psychology, and medicine than economics (for reviews: Welsh (1999); McDonald (2012); Fitzgerald and Cortina (2018)). Overall, the largest body of empirical evidence focuses on recording the prevalence of workplace sexual harassment. Secondly, the literature has established a link between workplace sexual harassment and career interruptions. Only recently has administrative data been used to study the consequences of workplace sexual harassment for worker sorting. Other work considers how reports of workplace sexual harassment may affect the firm (Abeysekera and Fernando, 2024). We build on this empirical work and document the first direct evidence of the earnings consequences of workplace sexual harassment. We also document that clients are an important source of workplace sexual harassment and consider possible spillovers to other workers.

While much is known about the prevalence of workplace sexual harassment, gaps remain in our understanding of its labor market consequences. Previous research finds, as does this paper, that sexual harassment from coworkers is more common when a worker belongs to a gender minority (Gutek and Cohen, 1987; Gutek et al., 1990; Gruber, 1998; Jackson and Newman, 2004; Kabat-Farr and Cortina, 2014; Folke and Rickne, 2022). Workplace sexual harassment also tends to occur more frequently in certain industries and occupations. However, much of this early research relies on cross-sectional survey data, which limits the ability to track workers over time and assess the long-term effects of harassment. The administrative panel data used in this paper allows for a more comprehensive study of the dynamic labor market consequences following sexual harassment in the workplace not only for victims, but for firms as well.

Previous research has also established the significant impact of workplace sexual harassment on workers' labor market mobility and career disruptions (Neumark and McLennan, 1995; Feldblum and Lipnic, 2016). Early studies attempt to measure mobility by examining intentions to quit through survey data (Laband and Lentz, 1998; Antecol and Cobb-Clark, 2004; Willness et al., 2007). The strongest evidence in the U.S. context comes from a panel study of young women in St. Paul, Minnesota, which finds that workplace sexual harassment is associated with increased financial stress and job moves (McLaughlin et al., 2017). More recently, Folke and Rickne (2022) provides the first evidence on career interruptions using administrative data, showing that mobility rises following incidents of workplace sexual harassment. Looking more broadly to examine other forms of workplace violence, Adams-Prassl et al. (2024) found that victims of workplace violence experience declines in employment rates.

Our main empirical contribution is to offer the first estimates of the magnitude of earnings losses following workplace sexual harassment. Prior research has tried to establish whether there exists an effect of workplace harassment on wages and earnings, but has been limited by data constraints. For instance, (Hersch, 2011) attempts to calculate the cost of workplace sexual harassment from a legal perspective when seeking damages using limited data from reports to the EEOC. While they do not observe earnings directly, Folke and Rickne (2022) find that workers move to more gender-segregated and lower-paying workplaces following workplace sexual harassment. They also performed a survey to illicit workers' willingness to pay to avoid workplace sexual harassment. Here they find that workers are willing to forgo 10 percent of their wages to avoid a workplace where they'll be exposed to workplace sexual harassment, which is consistent with the magnitude of earnings declines we observe from movers following harassment. Work by Batut et al. (2022) builds on these findings using administrative and survey data from France to document mobility outcomes following reports of workplace sexual harassment. Although they do not have information on earnings either, their results also provide evidence that workers who face exposure to workplace sexual harassment move to lower-paying firms. We offer novel direct evidence of the effect of workplace sexual harassment on earnings.

Another important area of research on workplace sexual harassment focuses on survey design. How questions about harassment are framed in surveys can significantly influence workers' willingness to disclose their experiences. Stockdale et al. (1995) and Ilies et al.

(2003) emphasize that the most effective approach is to focus specifically on behaviors that fall under the definition of workplace sexual harassment. Related work also explores the factors that influence workers' decisions to report incidents to their employer. Unfortunately, reports of harassment are sometimes ignored or, in the worst cases, met with retaliation, see Bergman et al. (2002); Cortina and Magley (2003), which discourages employees from coming forward. Recent evidence suggests that employer policies impact reporting behavior, with some policies even backfiring (Cheng and Hsiaw, 2022). Similarly, Dahl and Knepper (2022) documents the importance of workers' outside options in their decision regarding whether to report sexual harassment in the workplace.

In the remainder of this paper, Section 2 discusses the data and sample selection, section 3 documents our empirical methods, and Section 4 reports the results. And finally, Section 5 concludes.

2 Data

In this section we describe how we construct the novel dataset used int his paper that combines labor market information with instances of workplace sexual harassment. And, we present descriptive statistics that summarize important features of the micro data worth highlighting.

2.1 Data Sources and Sample Selection

In order the identify workers who are victims of workplace sexual harassment we use survey data from Denmark. The survey data come from the study 'Working environment and health in Denmark'. From 2012 until 2018, the National Research Center for the Working Environment has asked the same questions to different sets of employees in Denmark every two years. The purpose of the survey was the analyze how the employees themselves assess their physical and mental working environment as well as their health and ability to work. The survey asks respondents two questions that measure the frequency and sources of workplace sexual harassment.

We link this repeated cross-sectional data to employment histories and earnings from the

Danish employer-employee-matched administrative data. We focus our sample on workers from 2008-2021. This gives us individual labor market histories both before and after individuals respond to the workplace survey. For the key employment and earnings variables, we use the Integrated Database for Labor Market Research (IDA) register. We supplement this with additional registers to get information about education and other demographic variables (BEF, UDDA).

The main outcomes we consider are earnings and firm mobility. Earnings are the annual gross labor market income deflated to the 2015 price level.[2] To capture the effects of employment flows on earnings, we include periods of zero earnings. We also track whether the worker remains at the firm they were in when they were surveyed. To ensure that survey responses match the workers current employer, we only consider worker who have been at their place of employment for at least one year.[3].

2.2 Descriptive Statistics

We first analyze the demographic characteristics of the survey respondents. The main workplace environment survey question is "Have you been exposed to sexual harassment in the past 12 months?" where workers respond whether they were exposed daily, weekly, monthly, at least once, or never. Overall, 3 percent of the sample reported that they had been exposed to sexual harassment in the past 12 months. For women, the rate was 4.5 percent, and for men, it was 1.2 percent. This response rate is lower than other estimates for Denmark. An EU-wide survey on violence against women in the workplace estimated the 12-month incidence for women at 10 percent in Denmark. We attribute the discrepancy between our survey and other reports to differences in the subjective nature of the questions framing. The survey question we use relies on self-identification of harassment which is known to skew self-reports downward (Stockdale et al., 1995).

Table 1 reports demographic summary statistics for workers who responded that they had faced any amount of exposure to workplace sexual harassment. Note that the respondents

[2]Earnings outcomes are in 2015 USD.

[3]We explore employment outcomes as well, but find limited effects. See Appendix E

Have you been exposed to sexual harassment?

	% Women	Age	Ann. Earn (USD)	Tenure	College	Obs
Yes	82	42.4	52,012	5.21	.35	2,907
No	53	46.91	62,055	6.90	.41	99,555
Total	54	46.79	61,770	6.85	.41	102,462

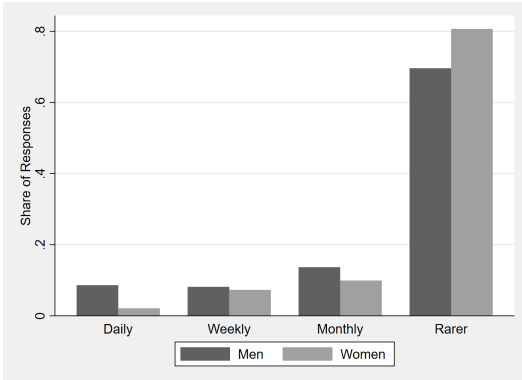
Table 1: Summary Statistics

Notes: This table shows summary statistics for workers for those exposed and those not exsposed to sexual harassment for the fraction of women, age, annual earnings which are the annual gross labor market income deflated to the 2015 price level, tenure measured in years and college which is an indicator variable. Number of observations in each category are in the final column.

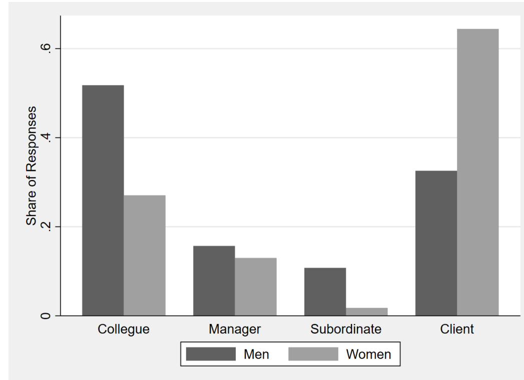
who report workplace sexual harassment are overwhelmingly women. They also tend to be younger, earn less, and have a lower rate of college attendance. We also document differences in reports of workplace sexual harassment by gender and age (See Appendix A). Over 10 percent of women under 25 in the sample report being exposed to workplace sexual harassment in the past year while around 3 percent of men in this age group reported an exposure. In Appendix B, we report demographic summary statistics for workers who responded that they had faced exposure to workplace sexual harassment by the frequency of the exposure and source of harassment.

Figure 1a decomposes the frequency of workplace sexual harassment for workers in the survey who report any degree of exposure. For most workers, sexual harassment is a rare event, although a significant number of respondents report repeated exposure to workplace sexual harassment. Again, there are slight differences by gender with men who report exposure to workplace sexual harassment typically reporting a higher frequency of incidences.

Finally, we can examine the sources of exposure to workplace sexual harassment in the data. The relevant survey question asks, "Who exposed you to sexual harassment?" In the survey, respondents who reported exposure to workplace sexual harassment had the option to indicate whether the source of the exposure was from colleagues, managers, subordinates, or clients/patients/etc. Respondents may report more than one source of sexual harassment.



(a) Frequency of sexual harassment



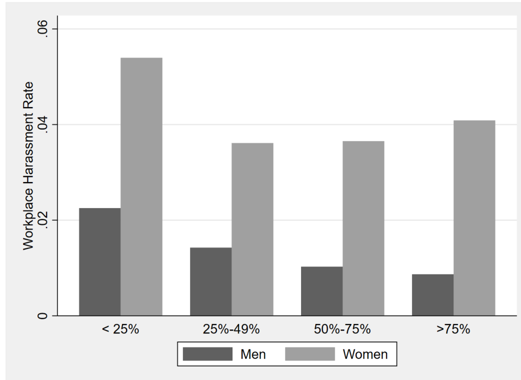
(b) Sources of sexual harassment

Figure 1: Frequency and Sources of workplace sexual harassment

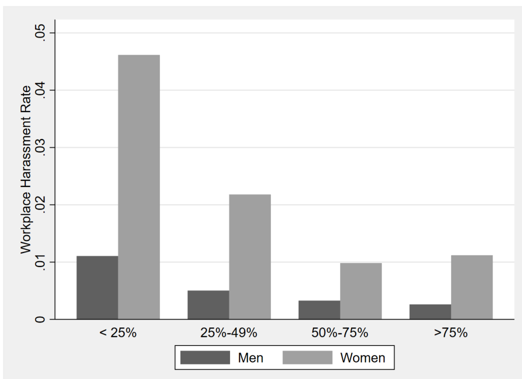
Notes: Both plots record the distribution of responses among male and female workers who report any degree of workplace sexual harassment. Panel (a) plots the reported frequency of workplace sexual harassment into daily, weekly, monthly or rarer and (b) plots the source of reported workplace sexual harassment, whether it is colleague, manager, subordinate or client.

Figure 1b reports the sources of workplace sexual harassment by gender. For both men and women, the most common sources of harassment were colleagues and clients. For men who indicated exposure to workplace sexual harassment, 50 percent of respondents reported colleagues as the source of exposure and 33 percent reported clients. For women, 27 percent reported colleagues as the source, and 64 percent reported clients. Men and women report a similar degree of exposure by managers, and men report a higher degree of exposure by subordinates.

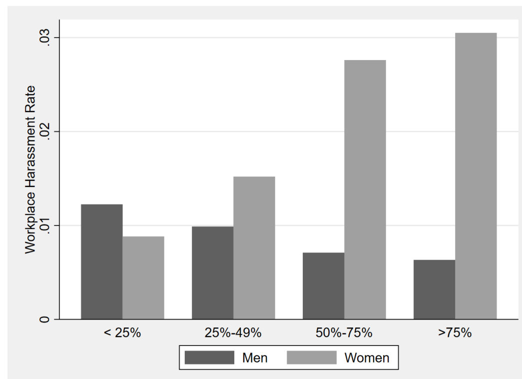
Prior work has emphasized the relationship between firm or occupation composition and workplace sexual harassment (Folke and Rickne, 2022). Motivated by these results, we look at reports of sexual harassment by the gender composition of the firm. We calculate the share of men in each firm and drop firms with less than 5 employees for this analysis. Figure 2a plots the results for all reports of workplace sexual harassment. At first, it appears that we do not see the same trends documented in prior work that risk is higher for gender minorities. While we do see this pattern for men, for women, we observe a U-shape in risk. Figure 2b plots the results by gender and separately for instances due to clients and Figure 2c does the same for colleagues. We see that for men and women, the incidence of reports of workplace



(a) All harassment



(b) Client harassment



(c) Coworker harassment

Figure 2: Sources of workplace harassment by firm male share and gender

Notes: Panel (a) plots the rates of reported workplace sexual harassment stemming from any source by gender depending on the firm's male employee share (on the first axis) being below 25 percent, 25 percent to 49 percent, 50 to 75 percent of above. Panel (b) plots the rates of reported workplace sexual harassment stemming from clients, patients, or students by gender depending on firm male shares. Panel (c) plots the rates of reported workplace sexual harassment by gender stemming from coworkers - either a colleague, manager, or subordinate depending on firm male shares.

sexual harassment from coworkers increases in the share of the opposite gender. However, we also document that when it comes to client-based harassment the rate is higher for both men in women in workplaces with a higher female share. This client-based harassment drives the bottom end of the U-shape in harassment for women.

Finally, we more closely examine differences in workplace sexual harassment across occupation categories. Table 2 reports harassment rates by gender and occupation group. We also include average pay by gender in these occupations. Women in agricultural and service

Workplace Sexual Harassment by Occupation

Occupation	% Women	Ann. Earn (USD)	% Women Harassed	% Men Harassed	Obs
Agriculture	9	64,097	8.9	0.8	540
Managers	32	106,784	2.1	0.7	4,514
Professionals	63	69,792	3.7	0.9	29,840
Technicians	58	64,996	2.5	0.8	12,439
Administrative	73	52,254	2.6	1.0	8,717
Service	72	44,600	9.0	2.4	12,363
Crafts	7	57,064	3.3	0.7	6,833
Assembly	21	54,785	3.6	1.0	4,904
Manual	43	44,827	2.4	1.6	5,848

Table 2: Harassment Rates and Summary Statistics by Occupation

Notes: This table presents the statistics for workers for those exposed in each occupation, where we use broad DISCO codes to characterize occupations. The first column is the fraction of women, the next is annual earnings which are the annual gross labor market income deflated to the 2015 price level, then we report the percentage of women harassed and men. The number of observations is each occupation are stated in the final column.

occupations face the highest rate of harassment with 9 percent of women reporting exposure. We use broad DISCO codes to characterize occupations.[4] Service occupations tend to have high rates of harassment for both men and women and the lowest pay. However, women comprise over three-quarters of workers in service occupations. The sources of sexual harassment in the workplace also vary by occupation, with women sorting more heavily into occupations that are exposed to high rates of sexual harassment by clients than men, while other occupations see a greater gender divergence in workplace sexual harassment stemming from coworkers.

[4] Six-digit DISCO codes are used for classifying and aggregating information about job functions in clearly defined groups, in relation to the tasks performed in the job or occupation.

3 Identification and Estimation

Next, we estimate the impact of workplace sexual harassment on labor market outcomes. To do this, we use a multiple-period difference-in-difference approach to analyze the outcomes of workers who report workplace sexual harassment. We considered different years of the survey to be the different periods of the treatment. We follow the approach of Callaway and Sant'Anna (2021) in constructing our estimator in which we compare the outcomes for workers who report harassment to those who never report workplace sexual harassment.

We are interested in several key labor market outcomes of workers who experience sexual harassment on the job. We focus mainly on the log annual labor earnings and job mobility, which we define as an indicator equal to one if the worker is at a firm different from the one that they were at in the year they responded to the survey.

The staggered difference-in-difference estimation method we use relies on several assumptions (Callaway and Sant'Anna, 2021). The first assumption is the irreversibility of the treatment such that once a unit is treated that treatment status is permanent. This implies that once a report of workplace sexual harassment occurs, the record of that event is permanent from the point of view of the worker. The second assumption is random sampling, which implies that each unit is randomly drawn from a large population of interest. The third assumption is limited treatment anticipation. For this to hold, we assume that individuals do not anticipate their experience of workplace sexual harassment prior of it occurring. And finally, we make the conditional parallel trends assumption based on the "Never treated" groups such that conditional on the large set of controls including demographics and employment characteristics, the treated worker's labor outcomes would evolve in parallel to those who we do not observe an experience of workplace sexual harassment for (Callaway and Sant'Anna, 2021).[5] [6]

As in the procedure outlined in Callaway and Sant'Anna (2021), we start by estimating

[5]Our never treated group is defined from the point of view of what we observe in the data. While this group of workers does not report sexual harassment in the survey, they may indeed experience sexual harassment outside of the observed survey waves or not report the experience.

[6]We further examine the validity of the conditional parallel trends assumption in Appendix C.

the group-time average treatment effect on the treated *ATT(g,t)*:

$$ATT(g,t) = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1] \tag{1}$$

where t is the time-varying indicator, G is the time period when a unit first becomes treated, and G_g is a binary variable equal to one if a unit is first treated in period g. We use the improved doubly-robust estimation procedure based on the inverse probability of tilting and weighted least squares from Sant'Anna and Zhao (2020) to estimate the $ATT(g,t)$. Given the differences we see in which demographic groups report selection into workplace sexual harassment, we include a vector of covariates. The covariates we include are education, occupation, gender, age, and age squared. We also restricted the sample to individuals who have been at their firm for over a year to ensure that their responses correspond with their firm at the time of the survey.

We are ultimately interested in aggregating these estimated effects across groups and time. The main focus of our results is the simple weighted average of all $ATT(g,t)$'s.

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} w(g,t) ATT(g,t) \tag{2}$$

where $\mathcal{G}$ is the set of groups and $w(g,t)$ is a weighting function. We use the simple weights from Callaway and Sant'Anna (2021) which creates a weighted average of the $ATT(g,t)$'s, where the weights sum to one (avoiding any negative weight issues) and puts more weights on larger units.

4 Results

In this section, we examine whether workers who have experienced sexual harassment earn less, whether they move to other firms, and we finally investigate the mechanisms behind the results.

Individual Earnings Losses Following Sexual Harassment

	(1)	(2)
	Women	Men
Coworkers & Clients	-0.0580***	-0.0636***
	[-0.0738,-0.0423]	[-0.0943,-0.0329]
Clients	-0.0482***	-0.0819*
	[-0.0661,-0.0303]	[-0.151,-0.0125]
Coworkers	-0.0706***	-0.0644***
	[-0.102,-0.0395]	[-0.102,-0.0272]

95% confidence intervals in brackets

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3: Estimated earnings losses following workplace sexual harassment by source

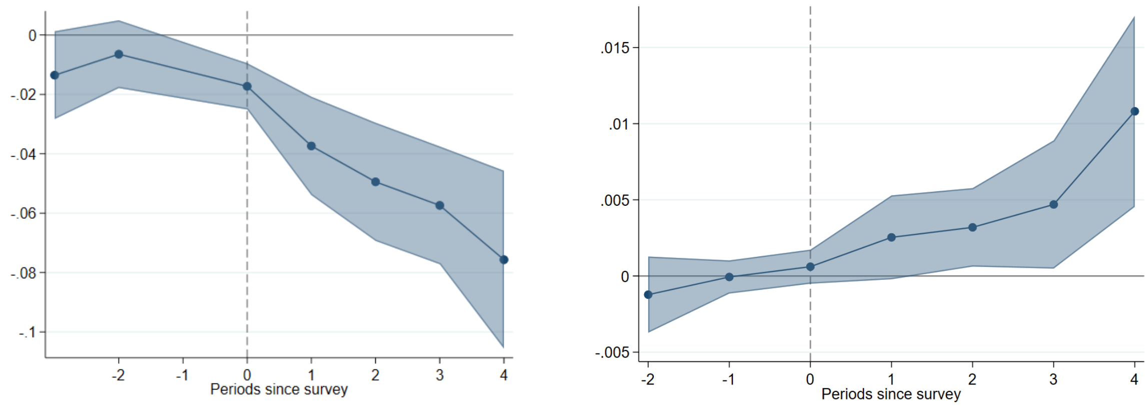
Notes: This table reports Individual earning losses following sexual harassment for coworkers and clients. Coefficient estimates are expressed as a percentage of earnings. Estimates are derived using a multiple-period difference-in-difference approach. 95 percent confidence intervals and p-values are obtained following Callaway and Sant'Anna (2021).

4.1 Earnings

Table 3 reports the estimated static aggregate ATT for earnings. The first row reports the ATT for earnings with all sources of workplace sexual harassment for women and men. Women see a 5.8 percent decline in earnings and men see a 6.3 percent decline. We also estimate the earnings results separately for exposure that comes from coworkers and exposure that comes from clients. The second row reports the estimated ATT for client-based harassment and row three reports the estimates for coworker-based harassment. The biggest gap in earnings effects between men and women comes from client based harassment, with men's earnings losses at almost twice the magnitude of women's losses. Women on the other hand, see slightly larger losses following harassment by a coworker than men.

4.2 Mobility

Next, we investigate whether victims "escape" workplace sexual harassment after they report exposure by leaving their job. To measure job mobility, we record an indicator for every



(a) Earnings

(b) Mobility

Figure 3: Dynamic ATT for earnings and mobility

Notes: Panel (a) plots the estimates for the dynamic ATT aggregator for earnings (b) plots the estimates for the dynamic ATT aggregator for firm departures. Both samples include men and women and all sources of workplace sexual harassment.

period that the worker is observed at an employer that differs from the one they were at in the survey year. We then construct the aggregate dynamic weighted average of the $ATT(g,t)$'s, which provides the average effect of participating in the "treatment" for the group of units that have been exposed to the treatment for exactly e periods. This analysis is similar to the standard two-way fixed effect regression in terms of outcomes of interest but avoids the issues with negative weighting that were been highlighted by Goodman-Bacon (2021). The dynamic aggregator gives us a sense of the dynamic impact of workplace sexual harassment and allows us to examine the conditional parallel trends assumption. We present the results for earnings and for mobility in the main sample with all sources of workplace sexual harassment for both men and women in Figure 3.

Appendix C reports the results of the parallel trend tests for the static ATT for firm and worker earnings, as well as for the firm male share. This is an additional test of our assumption of parallel trends in relation to the never-treated units.

Individual Earnings Losses by Job Mobility

	(1)	(2)
	Women	Men
Movers	-0.138***	-0.175***
	[-0.181,-0.0955]	[-0.251,-0.0980]
Stayers	-0.0284***	-0.0235
	[-0.0423,-0.0144]	[-0.0563,0.00919]

95% confidence intervals in brackets

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: Estimated ATT for earnings for movers vs. stayers by gender.

Note: This table reports estimated earnings losses for those who stay in a firm and those who leave after the incidence of sexual harassment. Coefficient estimates are expressed as a percentage of earnings. Estimates are derived using a multiple-period difference-in-difference approach. 95 percent confidence intervals and p-values are obtained following Callaway and Sant'Anna (2021).

4.3 Mechanisms

So far, we have established that victims of workplace sexual harassment experience both earnings losses and also leave their firms at higher rates. Next, we examine the characteristics and consequences of these moves for the labor market outcomes of victims.

First, we split the sample and estimate the specification from 1 on movers and stayers separately. We define movers as individuals who are employed at a different firm than one reported in the survey three years after the survey. Table 4 reports the results of the simple static ATT. We see that earnings losses are concentrated among workers who leave their firms. Column (1) reports the results for women and column (2) reports the results for men. We see that average earnings losses are driven by the larger magnitude of losses for workers who move jobs. Men who move also have larger earnings losses than women who move, which are 17.5 percent and 13.8 percent, respectively.

Given the significance of mobility, we further examine possible sources of losses following these transitions. First, we analyze occupational mobility outcomes, where we define occupational mobility as an indicator equal to one when the worker is employed in an occupation different from the one they reported in the survey year. This classification is determined

Occupational Mobility & Firm Characteristics

	(1)	(2)
	Women	Men
Occupation Moves	-0.0110	-0.0358*
	[-0.0249,0.00296]	[-0.0704,-0.00114]
Firm Size	-0.0559	0.0140
	[-0.126,0.0139]	[-0.150,0.178]
Firm Male Share	-0.00718**	0.0182**
	[-0.0115,-0.00290]	[0.00652,0.0299]

95% confidence intervals in brackets

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: Estimated ATT for occupation and firm outcomes

Note: This table reports estimated occupational mobility related to firm characteristics, such as firm size and firm male share. Coefficient estimates are expressed as a percentage of the respective outcome variable. Estimates are derived using a multiple-period difference-in-difference approach. 95 percent confidence intervals and p-values are obtained following Callaway and Sant'Anna (2021).

at the six-digit occupational code level. Table 5 reports the results for the different firm characteristics we examine by gender. The first row reports the results for occupational moves. This analysis shows that occupational mobility is not higher for women exposed to workplace sexual harassment and is, in fact, lower for men in the three years following harassment. This finding suggests that losses in occupation-specific human capital are not a significant factor driving the earnings losses we document in Section 4.1.

We also analyze the role of workplace sexual harassment in how workers sort into firms with varying levels of productivity. Previous research suggests that women subject to workplace sexual harassment may transition to lower-productivity or lower-paying firms. As a measure firm productivity, we use firm size, defined as the number of full-time employees. The second row of Table 5 shows that we fail to reject the null for both men and women. Again, this suggests that firm productivity does not play a substantial role in driving losses experienced by movers.

Finally, we also consider the gender composition of the firms that workers work at before and after reports of workplace sexual harassment. Row three reports the impact on the male share of the workplaces of victims of sexual harassment. Consistent with prior work, we see that women who report workplace sexual harassment move to firms with fewer men. Furthermore, we document that men move to firms with fewer women. This suggests that preferences for increased gender segregation may be the key channel by which workers face earnings losses following workplace sexual harassment. Men and women may accept lower wages to work in a workplace with a greater proportion of colleagues who share their gender.

4.4 Coworker Spillovers

To examine the spillovers of workplace sexual harassment on earnings for individuals who work at a firm that is exposed to sexual harassment, we conduct an analogous event study. We measure the effect of a workplace with sexual harassment on average firm earnings and the effect of a harassment event on non-harassed workers' average earnings.

Let $Y_{s,t}$ be the average log annual earnings at the firm level in year t for gender s. In Table 6 we estimate equation (2) for all employees at a firm and separately for those who we cannot identify experiences of sexual harassment i.e. the "non-harassed".

Since we do not observe the universe of workers in our survey data, misclassification errors may be present in which firms have harassment incidents that are unobserved. Thus, we take these estimates as a lower bound on the earnings spillover of workplace sexual harassment. Column 1 and 2 of Table 6 report the earnings growth for women and men at a firm following a sexual harassment event, respectively. We find that both the overall firm spillover the spillover for those who have not experienced harassment is near zero. For all male coworkers in a firm and for the "non-harassed", earnings grow by about one percent. We find no evidence of changes in female coworkers' subsequent earnings growth at firms that experience a sexual harassment incident relative to those that do not. This may be due to the fact that women are, in general, more likely to experience sexual harassment, and these types of events are therefore considered in alignment with the norm for women. Overall, we find little effect of workplace sexual harassment instances on coworker earnings spillovers.

Firm-level Earnings Following Sexual Harassment

	(1)	(2)
	Women	Men
Firm Spillover	0.00177	0.009057**
	[-0.00454, 0.00808]	[0.00266, 0.01545]
Non-harassed Spillover	0.00195	0.00901**
	[-0.00436, 0.00827]	[0.00260, 0.01542]

95% confidence intervals in brackets

* $p < 0.05$,** $p < 0.01$, *** $p < 0.001$

Table 6: Estimated avg. earnings growth at the firm following workplace sexual harassment by gender

Notes: This table reports estimated spillover effects on earnings of witnesses to workplace sexual harassment. Coefficient estimates are expressed as a percentage of the respective outcome variable. Estimates are derived using a multiple-period difference-in-difference approach. 95 percent confidence intervals and p-values are obtained following Callaway and Sant'Anna (2021).

4.5 Validity of Research Design

For robustness, we can perform a placebo test that creates fake indicators for experiences of workplace sexual harassment. We generate artificial experiences of workplace sexual harassment in the data in proportion to the actual experiences of workplace harassment we observe and allocate them to observations randomly. This allows us to estimate aggregate ATTs for earnings which we would expect to be approximately zero given that they were randomly assigned. The concern is that workers who experience sexual harassment are more likely to also experience a shock to their earnings growth in the same period. The results of this exercise in Figure 4 confirm that in our sample, the event study results are not driven by selection on-trend.[7]

[7] Appendix F reports the static aggregate gender-specific ATTs of the placebo test.

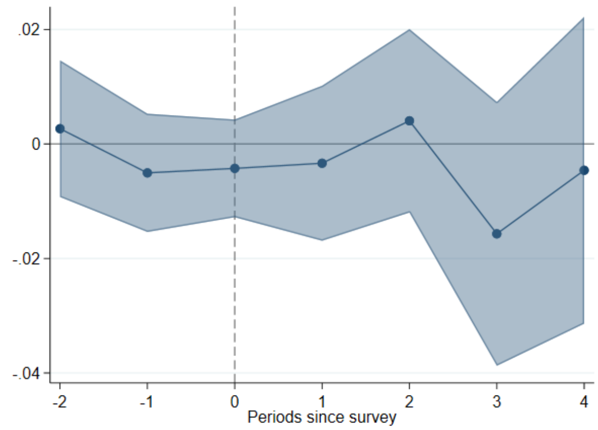


Figure 4: Dynamic ATT for earnings with Placebo Harassment

Notes: The dynamic ATT aggregator for log annual earnings using artificial indicators for experiences of workplace sexual harassment for both men and women.

5 Conclusion

The #metoo movement, which grew in prominence in 2017, put a greater spotlight on incidences of workplace sexual harassment. However, economic evidence has only just begun to document the extent of the labor market consequences of workplace sexual harassment.

In this paper, our main contribution is that we are the first to estimate the earnings consequences of these events. Using detailed survey and administrative data from Denmark, we can follow workers labor market histories after a report of exposure to workplace sexual harassment. These earnings losses are significant, and average 6 percent for both women and men in the four years following exposure. Looking at the dynamics of these losses, we see no evidence that workers earnings recover, as losses are driven by workers switching to new jobs.

We are also able to provide direct evidence of who perpetuates workplace sexual harassment and analyze differences in the incidence and consequences of these sources by gender. Prior work has focused on the role of coworkers as a source of harassment, especially for workers in a gender minority group. We do find coworkers to be an important source of harassment, but we also find that clients are the source for significant proportion of incidences as well, and make up the majority of incidences for women. Our empirical results show that the earnings consequences of harassment are of a similar magnitude for coworker- and client-based harassment. The sources of workplace sexual harassment should be a key consideration when developing workplace or government policies to address harassment.

Our work is part of a growing literature highlighting the economic implications of workplace sexual harassment and other forms of gender-based violence. We add more evidence to this work on the gender inequality of who faces workplace sexual harassment and the labor market consequences they face. Our findings can also inspire future work examining gender differences in workplace sexual harassment and the role of policy in addressing these inequalities on a larger scale.

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A Heterogeneity of Workplace Sexual Harassment

Figure 5 plots the percent of workers who reported any exposure to workplace sexual harassment in the past year by age and gender. We bin workers by age to meet release criteria since some age-gender groups have a small number of reports.

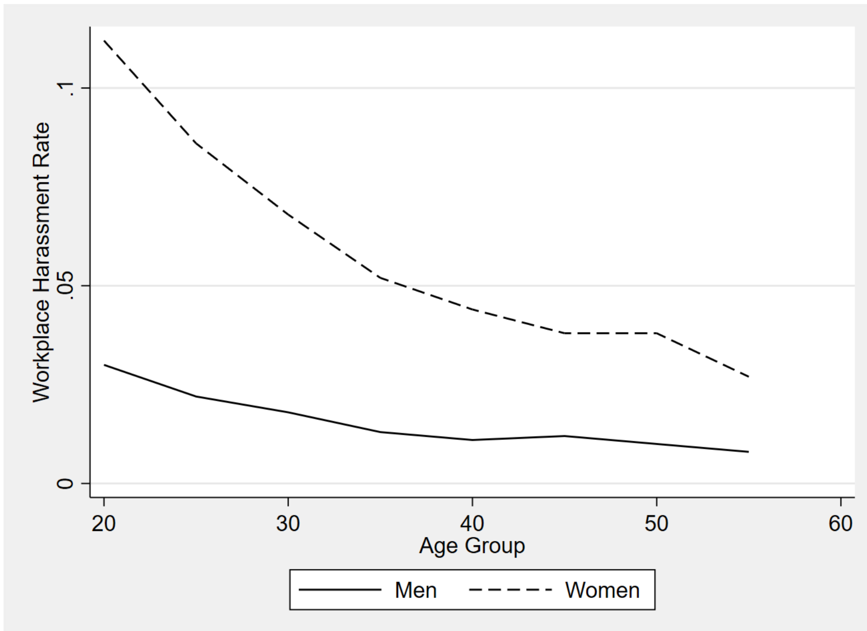

Figure 5: Incidence of harassment by age group and gender. Source: Author's calculations from AH data

B Descriptive Statistics Workplace Sexual Harassment

Table 7 displays the analysis sample's reported frequency of workplace sexual harassment incidents by worker characteristics. Annual earnings are the annual gross labor market income deflated to the 2015 price level, tenure is measured in years and college is an indicator variable. Number of observations in each category are in the final column.

Table 7: Summary Statistics by Harassment Frequency

	% Women	Age	Ann.Earn (USD)	Tenure	College	Obs
Daily	0.52	40.63	50,379	5.08	0.22	112
Weekly	0.80	40.45	48,007	4.02	0.19	256
Monthly	0.76	38.31	49,354	4.54	0.29	365
At least once	0.83	42.27	52,940	5.14	0.38	2,699
Never	0.53	46.57	62,482	6.72	0.41	112,595
Total	0.54	46.42	62,176	6.68	0.41	116,027

Table 8 displays the reported source of workplace sexual harassment incidents by worker characteristics for victims in our analysis sample. Annual earnings are the annual gross labor market income deflated to the 2015 price level, tenure is measured in years and college is an indicator variable.

Table 8: Summary Statistics by Harassment Source

	% Women	Age	Ann.Earn (USD)	Tenure	College	Obs
Client, Patient, etc.	0.90	42.77	49,394	4.79	0.32	2,003
Colleague	0.69	40.14	54,873	5.46	0.37	872
Manager	0.79	39.42	53,994	5.00	0.43	427
Subordinate	0.42	41.41	76,463	5.03	0.44	101
Total	0.82	41.64	52,159	5.00	0.35	3,403

C Parallel Trends

The average coefficient from our parallel trends test of pre-trends for various firm and individual outcomes are reported in Table 9.

Pre-Trends Test for Parallel Trends

	Average Pre-Trend Coefficient
Avg. Firm Wage	1.10e-41
Avg. Individual Wage	0.00086
Firm Male Share	0.0351

Table 9: Average pre-trends test on firm and worker characteristics

D Worker Mobility

Table 10 provides the estimated aggregate ATT for firm departures following incidents of workplace sexual harassment firm all sources by gender. We find that women are more likely to leave their workplace after experiencing sexual harassment at work than men.

Mobility Following Workplace Sexual Harassment

	(1)	(2)
	Women	Men
Coworkers & Clients	0.00845***	0.00210
	[0.00368,0.0132]	[-0.000486,0.00468]

95% confidence intervals in brackets

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 10: Estimated ATT for firm departures following workplace sexual harassment

E Employment

Table 11 provides the estimated aggregate ATT for an annual employment indicator to examine whether workers who experience sexual harassment at work are choosing not to participate in the labor force. However, we do not find evidence that these workers move into non-employment following such events.

Employment Following Workplace Sexual Harassment

	(1)	(2)
	Women	Men
Coworkers & Clients	0.0000	-0.0002
	[-0.0000,0.0000]	[-0.0007,0.0002]
Clients	0.0000	0.0000*
	[-0.0001,0.0000]	[0.0000,0.0000]
Coworkers	0.0000	-0.0006
	[-0.0000,0.0000]	[-0.0020,0.0007]

95% confidence intervals in brackets

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 11: Estimated ATT for individual annual employment following workplace sexual harassment

F Robustness

Table 12 reports the estimated aggregate ATT on earnings using equation 2 and artificial indicators of workplace sexual harassment. The estimated aggregate ATT in the first column for women is not statistically significant and the 95% confidence interval includes zero. The second column for men indicates that some of the earnings losses for men in our event study could be partially driven by a correlation between the timing of shocks to earnings growth and experiences of sexual harassment.

Placebo Test of Artificial Sexual Harassment Experience

	(1)	(2)
	Women	Men
Placebo Sexual Harassment	0.00513	-0.01589
	[-0.01494, 0.02521]	[-0.03762, 0.00583]

95% confidence intervals in brackets

* $p < 0.05$,** $p < 0.01$, *** $p < 0.001$

Table 12: Estimated earnings losses following artificial experiences of workplace sexual harassment by gender